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Kuo et al.

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(54) **STICK-SHAPED PAPER CONTAINER**

(71) Applicants: **Golden Arrow Printing Technology Co., LTD.**, New Taipei (TW); **L'Oreal**, Paris (FR)

(72) Inventors: **Chien-Kuan Kuo**, New Taipei (TW); **Chun-Huang Huang**, New Taipei (TW); **Simon Nicholas Dutton**, Basking Ridge, NJ (US)

(73) Assignees: **Golden Arrow Printing Technology Co., LTD.**, New Taipei (TW); **L'Oreal**, Paris (FR)

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Related U.S. Application Data

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B65D 8/00 (2006.01)

A45D 40/00 (2006.01)

B65D 6/34 (2006.01)

(52) **U.S. Cl.**

CPC **B65D 9/02** (2013.01); **A45D 40/00** (2013.01); **B65D 9/34** (2013.01); **A45D 2040/0025** (2013.01); **A45D 2200/05** (2013.01)

(58) **Field of Classification Search**

CPC ... B65D 9/02; B65D 9/34; B65D 1/14; B65D 75/5888; B65D 3/262; B65D 2565/382; B65D 65/466; A45D 40/00; A45D 2040/0025; A45D 2200/05; D21J 3/04; Y02W 90/10
USPC 229/4.5, 101.1, 201; 206/385
See application file for complete search history.

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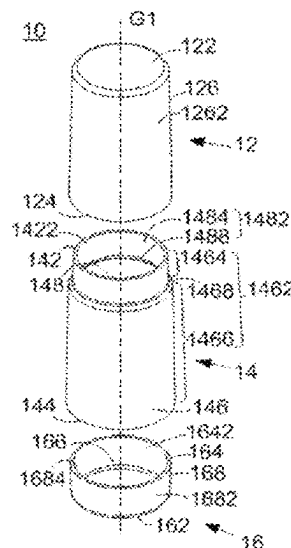
Primary Examiner — Christopher R Demeree

(74) *Attorney, Agent, or Firm* — CHRISTENSEN O'CONNOR JOHNSON KINDNESS PLLC

(57) **ABSTRACT**

A stick-shaped paper container according to the present invention is introduced herein, and comprises a covering body, a base seat and a sleeving unit, the respective entire structure of all which is integrally compression-molded with only rendering positive draft angles, respectively, only by a wet-fiber pulp-molding process applied for wet plant-fibrous pulps, thereby being capable of simplifying its mold assembly and lowering its mold cost. The covering body, the sleeving unit and the base seat are assembled together into the entire stick-shaped paper container having a container height-to-width ratio between 2~5.5.

13 Claims, 7 Drawing Sheets



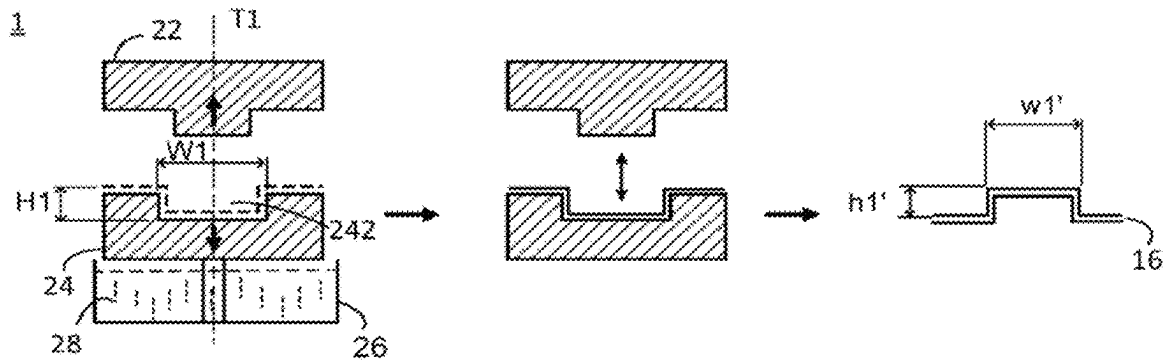


Fig. 1 (Prior Art)

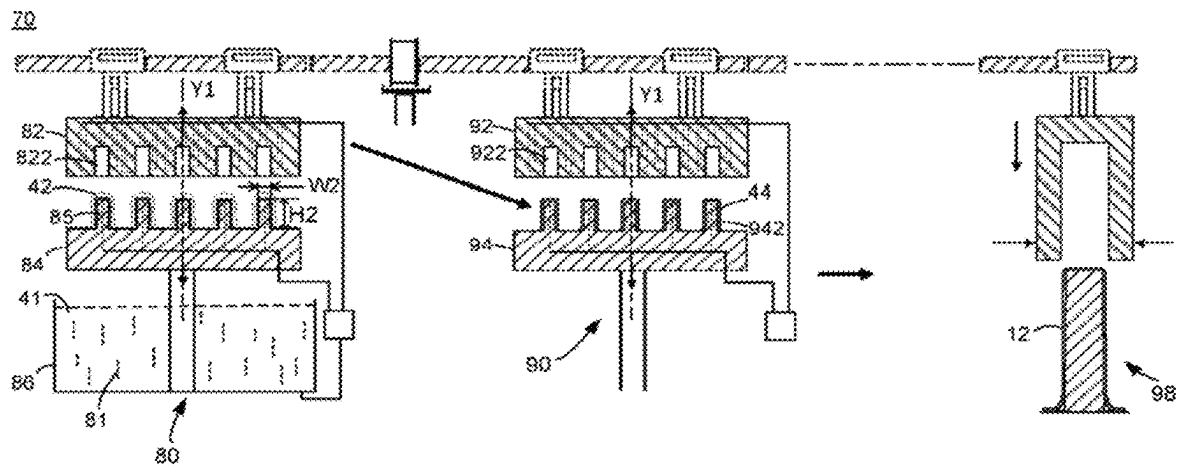


Fig. 2

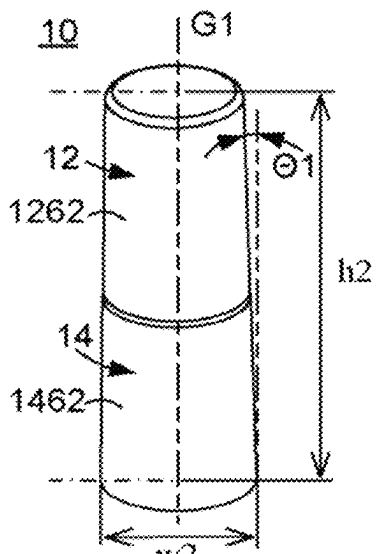


Fig. 3A

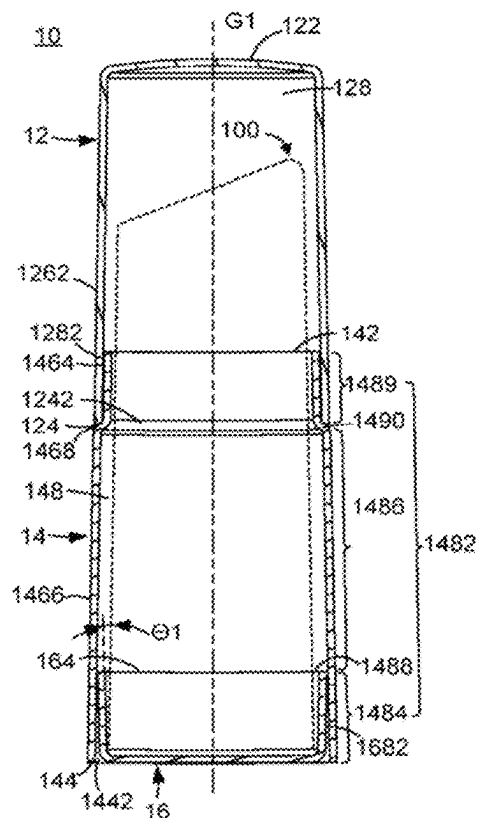


Fig. 3B

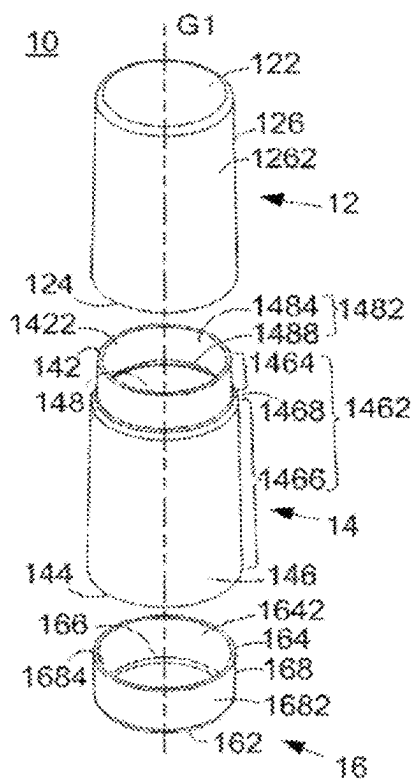


Fig. 3C

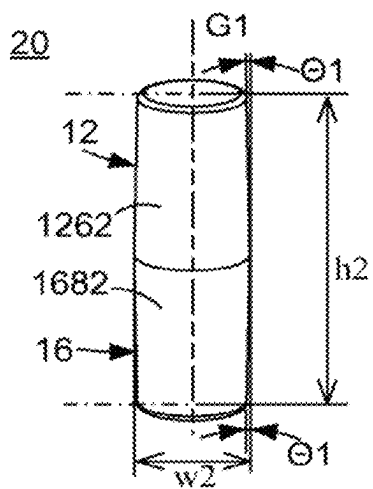


Fig. 4A

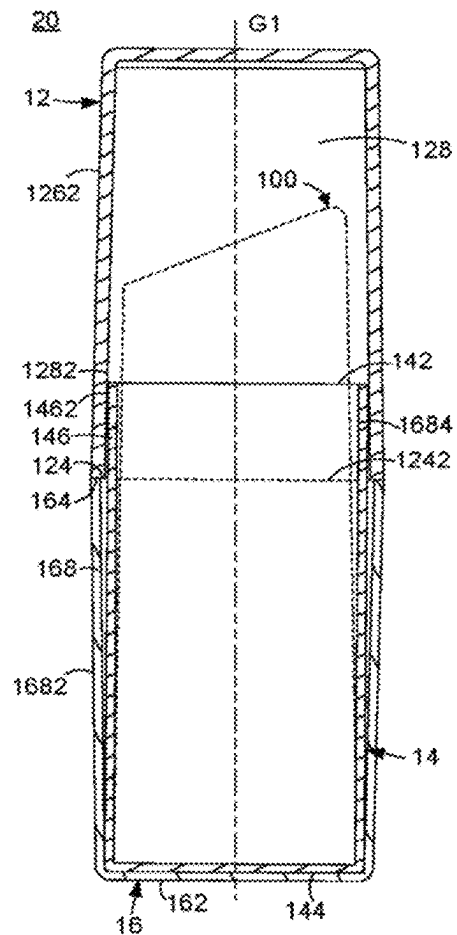


Fig. 4B

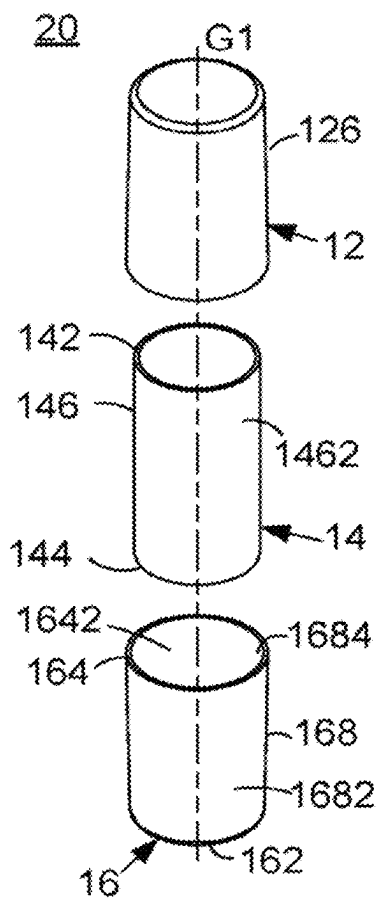


Fig. 4C

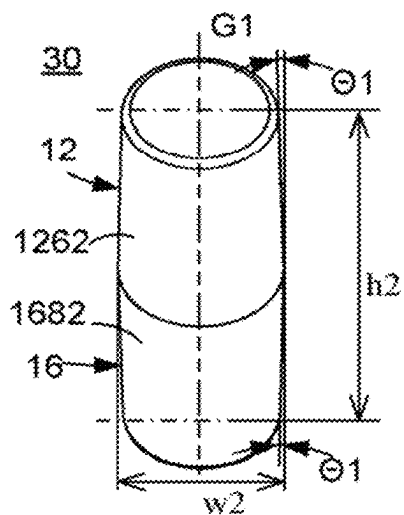


Fig. 5A

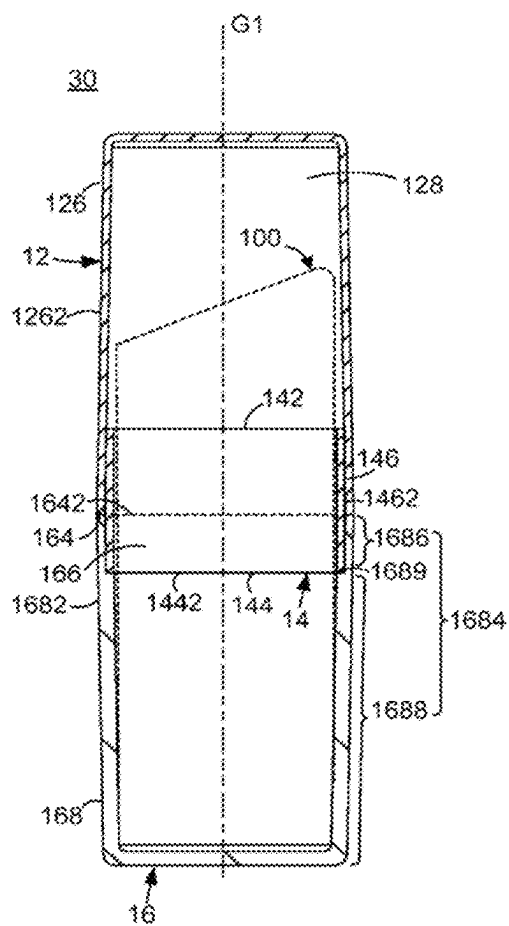


Fig. 5B

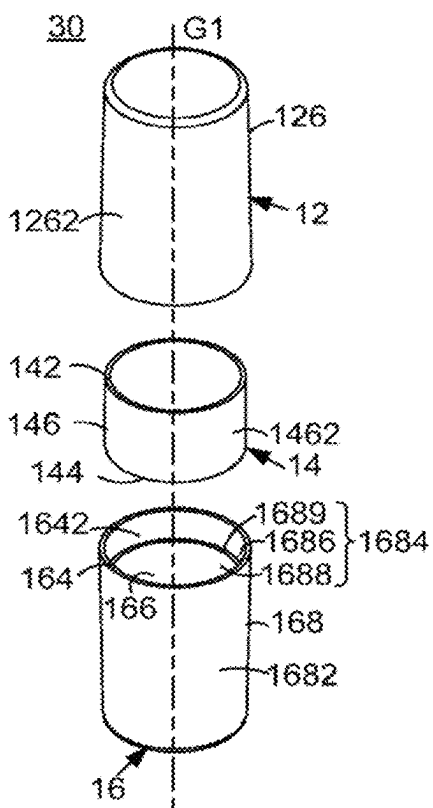


Fig. 5C

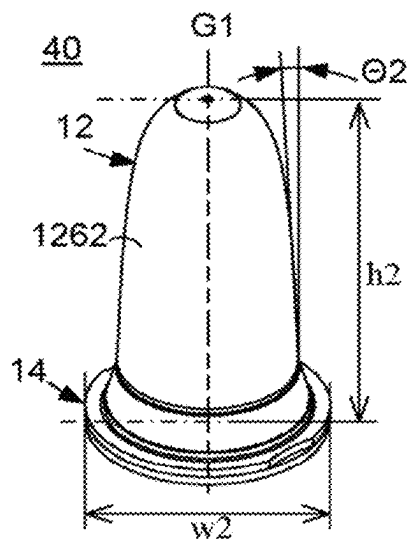


Fig. 6A

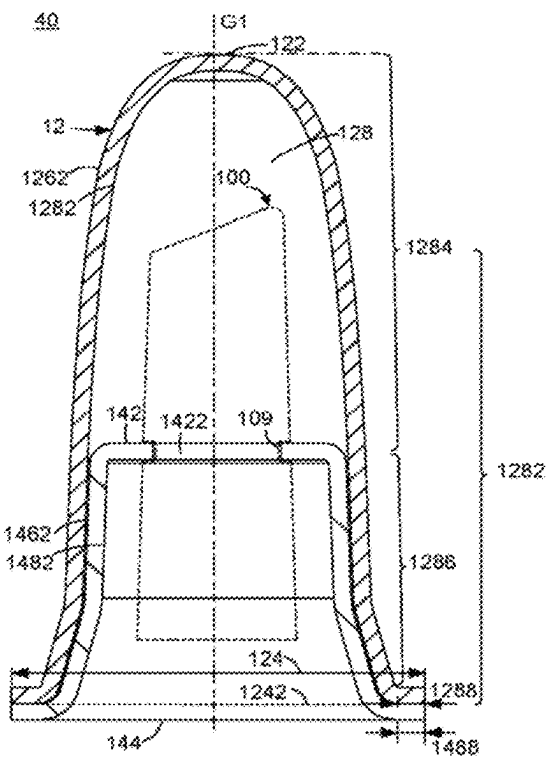


Fig. 6B

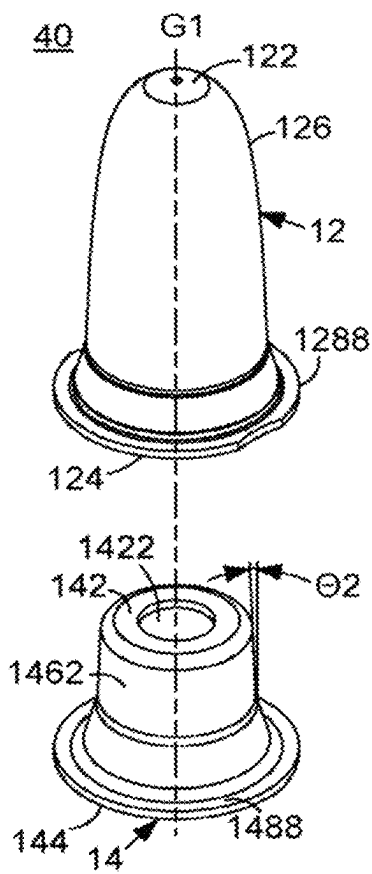


Fig. 6C

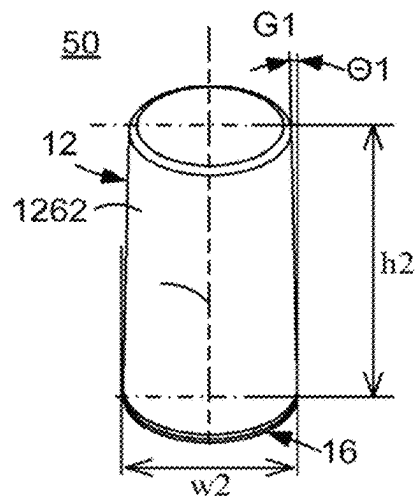


Fig. 7A

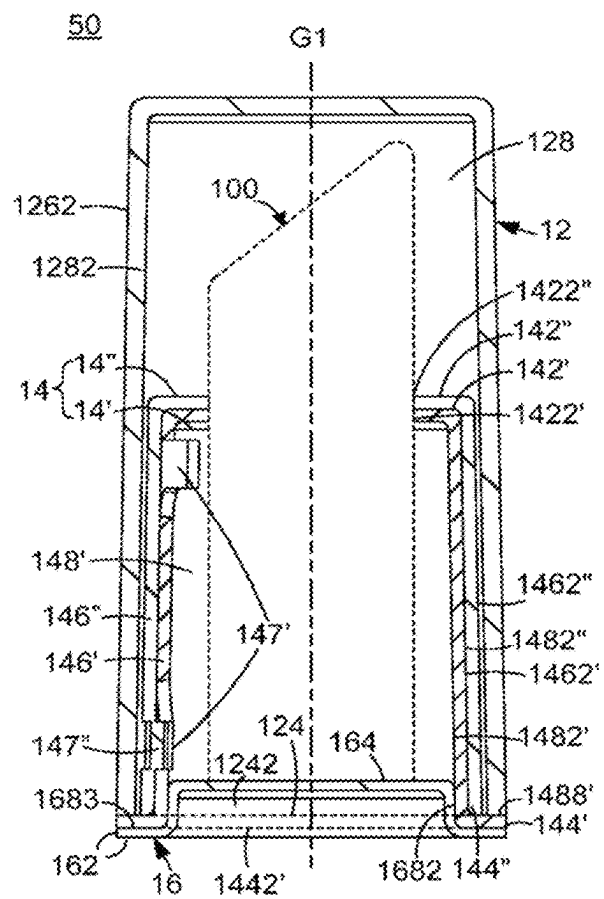


Fig. 7B

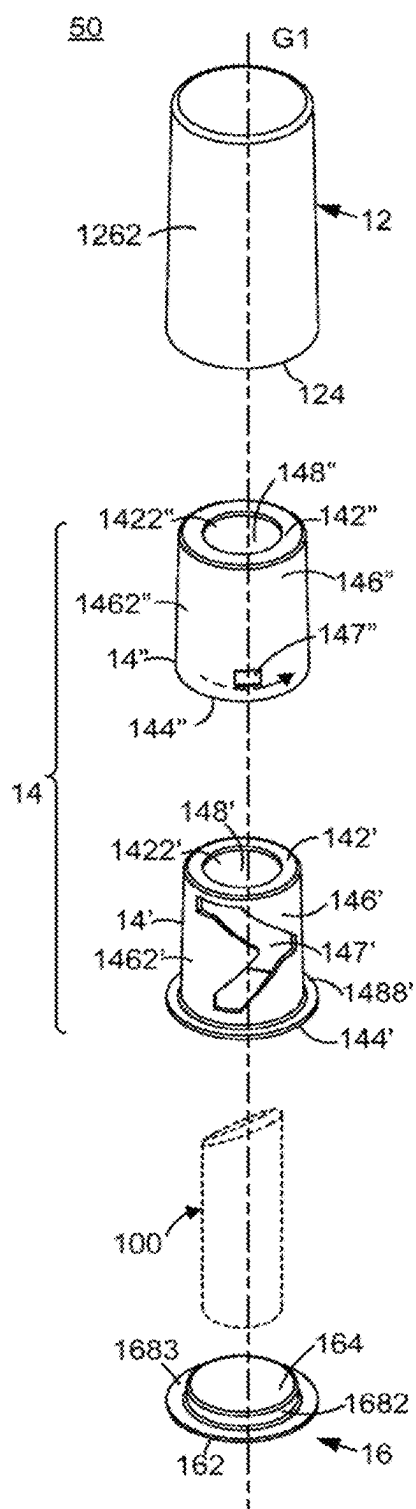


Fig. 7C

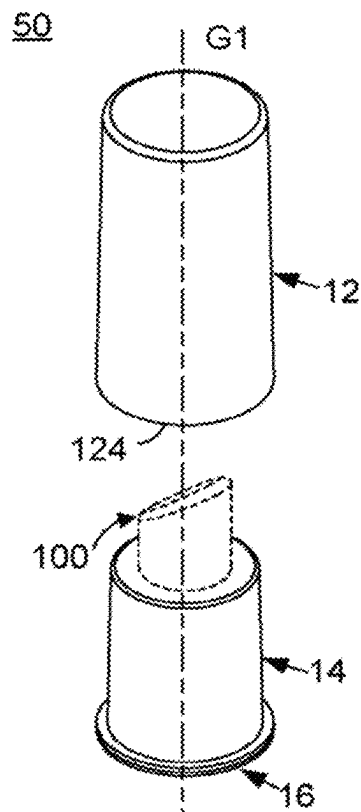


Fig. 7D

STICK-SHAPED PAPER CONTAINER

FIELD OF THE INVENTION

The present invention relates to a paper container applied in the technical field of a packaging material, and more particularly, is related to a stick-shaped paper container fabricated by a wet-fiber pulp-molding process.

BACKGROUND OF THE INVENTION

To speed up the manufacture for the conventional packaging container, mostly conventional packaging containers are integrally formed with several components thereof in plastic materials, or is assembled together with a few of paper parts and a few of plastic parts. This could not accomplish the latest environmental protection requirements for biodegradability or compostability; furthermore, if consumers' bodies, such as their lips or skins, frequently touch the plastic container (such as plastic packaging materials for lipsticks or cosmetics), it is liable to invoke healthy doubts for the human body, after a prolonged period. Presently, as depicted in FIG. 1, although a series of consistent production machines 1 where two coupled upper male mold 22 and lower female mold 24 are mutually compression-matched for wet plant-fibrous slurries positioned therebetween are deployed in a production line used for the conventional pulp-molding process (or so-called "wet-fiber pulp-molding process"), thereby integrally shaping a variety of huge 3C paper packaging materials from the slurries. Nevertheless, the conventional pulp-molding process exists with the following technical drawbacks: (1) during the conventional pulp-molding process after a dredging-pulp cave chamber 242 of the lower female mold 24 is configured to dredge up the wet plant-fibrous pulps 28 (or called "dredging-pulp") within from a slurry tank 26, then the lower female mold 24 and the upper male mold 22 both implement compression-molding together, a ratio R1 that a longitudinal molding depth H1 of the dredging-pulp cave chamber 242 of the lower female mold 24 is relative to a transversal molding width W1 of each side of the dredging-pulp cave chamber 242 mostly is restricted to be smaller than or equal to '1' (namely $H1/W1=R1$, $R1 \leq 1$). Through the upper male mold 22 and the lower female mold 24 both mutually compression-matching, a conventional pulp-molded article 16 is finalized in forming an integral shape which is frequently rendered in a compacted-and-flatted box body (as treated for a 3C packaging material) where its transversal width w1' is equal to or larger than its longitudinal height h1'; namely a ratio r1 of the conventional pulp-molded article 16 that the longitudinal height h1' (in parallel to the longitudinal molding depth H1) is relative to the transversal width w1' is smaller than or equal to '1' (namely $h1'/w1'=r1$, $r1 \leq 1$). If the transversal molding width W1 of the dredging-pulp cave chamber 242 of the lower female mold 24 is too small as well as the longitudinal molding depth H1 is too deep (namely its ratio R1 is larger than '1'), such a stick-shaped pulp-molded article possibly incurs a 'crosslinking effect' along a longitudinally molding aspect of the mold during its molding process. It represents that the molding process of the stick-shaped pulp-molded article will invoke its longitudinal sidewalls readily broken or having a too-thin thickness insufficient to provide a structural strength. This would lead to a drawback of a poor production yield.

Furthermore, while the stick-shaped pulp-molded article is fabricated by the conventional pulp-molding process, if the longitudinal sidewalls of the stick-shaped pulp-molded

article need to be molded together with laterally latch structure and/or laterally recessed structure thereon, various demold surfaces of the laterally latch structure and/or the laterally recessed structure would be rendered in negative draft angles with relation to two longitudinal demolding directions of both the upper male and lower female molds 22, 24, as collectively using a longitudinal centre line T1 (or so-called 'a vertical demolding direction'), whereas various demold surfaces of the other resting portions, such as its longitudinal sidewalls and its corresponding transversal bottom sidewalls, of the stick-shaped pulp-molded article are rendered in positive draft angles. This would invoke that after the article is being molded, the upper male and lower female molds both are longitudinally demolded as well as another transversal mold is horizontally demolded and/or a slidable block device is horizontally moved in cooperation therewith, thereby collaboratively molding the entire stick-shaped pulp-molded article which comprises the laterally latch structure and/or laterally recessed structure formed thereon. Understandingly, if the stick-shaped pulp-molded article has simultaneously different portions which are respectively rendered in both positive draft angles and negative draft angles, it would make the amount of the require mold components becoming increased and complex, for using with the conventional pulp-molding process. This will be bound to raise its mold device cost higher, make its repair difficulty growing, cause its production yield worse, and extend its cycle time of the production for the respective articles, simultaneously. Please refer to a PCT International Publication No. WO2020019098A1 that has been introduced, where an outer surface of an upper cover 10 is provided with a plurality of straight grooves 11 thereon, and an outer surface of a lower cover 20 is provided with a plurality of straight-and-large grooves 21 each formed with a step surface 23 on an inner side of an lower inner cavity 22 thereof, wherein the grooves 11, the straight-and-large grooves 21 and the step surface 23 (as a laterally latch structure) all that are laterally molded at negative draft angles, need to use another transversal mold implemented for a horizontally demolding, and/or a slidable block device implemented for a horizontal movement, in addition to cooperation together with the two male and female molds both implemented for the longitudinal demolding.

Hence, there is a demand to provide a stick-shaped paper container fabricated by an improved pulp-molding process, for resolving the afore-mentioned technical problems incurred in the conventional pulp-molding container made by the conventional pulp-molding process.

SUMMARY OF THE INVENTION

In order to solve a variety of technical matters incurred in the prior arts, a primary objective of the present invention is to provide a stick-shaped paper container, all components of which comprises, for example, a sleeving unit, a covering body, a base seat and so forth, and are capable of being integrally molded by a consistent-and-continuous production machine used with an improved pulp-molding process. This does not only lead to resolve a technical problem of the conventional pulp-molding process that is incapable of producing such a pulp-molded article which has a height-to-width ratio, greater than one, of its maximum longitudinal height being relative to its maximum transversal width, but also saves its working cycle time, benefits its mass production, and assures its higher product yield and quality.

Furthermore, another objective of the present invention is to provide a stick-shaped paper container, which adopts the

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pure plant fibers as materials for constituting the container, in order to avoid occurrence of the human healthy doubt and to conform with the FDA food-grade certification standard, thereby actually accomplishing an environmental protection requirement for both biodegradability and compostability.

Furthermore, another objective of the present invention is to provide a stick-shaped paper container, the components of which primarily comprises the sleeving unit, the covering body and the base seat, and have the respective demold surfaces all rendering only in positive draft angles relative to a longitudinal demolding direction of a compression-molding mold assembly used with the wet-fiber pulp-molding process, rather than the negative draft angles rendered in the laterally latch structure and/or laterally recessed structure of the conventional pulp-molded article are rendered. Accordingly, during the molding process, it would be unnecessary to employ a transversal mold implemented extra for a horizontally demolding, and/or a slidable block device implemented extra for a horizontal movement, thereby being capable of simplifying the mold assembly, and lowering its mold cost.

To accomplish the afore-mentioned objectives, a preferred embodiment of the present invention adopts the following technical solutions where a stick-shaped paper container, configured for accommodating a content therein, comprises: a sleeving unit, a covering body and a base seat.

The sleeving unit configured for restricting the content from transversal movement has a first and second end both opposed against each other, an annularly standing sleeve wall located between the first and second ends and formed around a longitudinal centre line, and a hollowed sleeve chamber that the longitudinal centre line passes through being formed between the first and second ends, wherein the first end is formed with a first opening space-communicated with the hollowed sleeve chamber and provided for the content passing therethrough, the hollowed sleeve chamber is formed with an innermost sleeve sidewall in a position of propping the content, and the standing sleeve wall is formed with an outermost sleeve sidewall.

The covering body has a closed top distal end, a fitting-on distal end opposed against the top distal end, a standing cover wall located between the top distal end and the fitting-on distal end and formed around the longitudinal centre line, and a hollowed cover chamber formed between the top distal end and the fitting-on distal end, wherein the fitting-on distal end is formed with a fitting-on opening space-communicated with the hollowed cover chamber, the hollowed cover chamber is formed with an innermost cover sidewall therein, the fitting-on distal end of the covering body is disposed at a position corresponding to the first end of the sleeving unit, and the standing cover wall has an outermost cover sidewall.

The base seat configured for longitudinally supporting the content has a closed bottom terminal, an abutting terminal opposed against the bottom terminal, a hollowed base chamber formed between the bottom terminal and the abutting terminal, and a standing base wall located between the bottom terminal and the abutting terminal and formed around the longitudinal centre line, wherein the abutting terminal is formed with a receiving opening space-communicated with the hollowed base chamber, the abutting terminal of the base seat is disposed at a position corresponding to the second end of the sleeving unit, thereby accommodating the content within between both the covering body and the base seat, and the standing base wall has an outermost base sidewall.

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Preferably, the covering body and the base seat both are integrally formed, respectively, only by a wet-fiber pulp-molding process where a male mold dredges up wet plant-fibrous pulps and then the male mold and a female mold both compression-mold the wet plant-fibrous pulps, in such a way that the entire structure of each of the covering body and the base seat all is formed with positive draft angles relative to the longitudinal centre line, wherein the covering body, the sleeving unit and the base seat are assembled together into the entire stick-shaped paper container which has a maximum container height parallel to the longitudinal centre line, a maximum container width perpendicular to the longitudinal centre line, and a container height-to-width ratio, between 2~5.5, of the maximum container height being relative to the maximum container width.

Preferably, the sleeving unit is integrally formed, by the wet-fiber pulp-molding process where the male mold dredges up the wet plant-fibrous pulps and then the male mold and the female mold both compression-mold the wet plant-fibrous pulps, in such a way that the entire structure of the sleeving unit is formed with positive draft angles relative to the longitudinal centre line.

Preferably, the outermost sleeve sidewall is divided, by along the longitudinal centre line, into an annularly outer narrow-diameter section, an annularly outer wide-diameter section, and an outer circular saddle formed on a boundary between the outer narrow-diameter section and the outer wide-diameter section. The first end of the sleeving unit passes through the fitting-on opening of the covering body to enter inside the hollowed cover chamber while the fitting-on distal end is disposed with the first end, wherein the innermost cover sidewall and the outer narrow-diameter section both are respectively formed with flatted surfaces longitudinally extended in such a way of constituting a plane-to-plane tight contact in a mutually resting-on manner between the innermost cover sidewall and the outer narrow-diameter section, and an inner diameter of the fitting-on opening is smaller than an outer diameter of the outer circular saddle, thereby blocking the fitting-on distal end on the outer circular saddle from moving.

Preferably, the second end of the sleeving unit formed with a second opening space-communicated with the hollowed sleeve chamber, and the innermost sleeve sidewall is divided, by along the longitudinal centre line, into an annularly-narrowed inner diameter section, an annularly-widened inner diameter section, and an inner circular saddle formed on a boundary between both the annularly-narrowed inner diameter section and the annularly-widened inner diameter, wherein the abutting terminal passes through the second opening of the sleeving unit to enter inside the hollowed sleeve chamber while the abutting terminal is disposed with the second end, the annularly-widened inner diameter section and the outermost base sidewall both have flatted surfaces extended toward the same direction in such a way of constituting a plane-to-plane tight contact in a mutually resting-on manner between the annularly-widened inner diameter section and the outermost base sidewall, and an outer diameter of the abutting terminal is smaller than an inner diameter of the inner circular saddle and smaller than the inner diameter of the fitting-on opening of the covering body, thereby blocking the abutting terminal on the inner circular saddle from moving.

Preferably, the sleeving unit is constituted by assembling at least two sleeves in a nested or stacked manner along the longitudinal centre line.

Preferably, while the at least two sleeves are assembled in the nested manner, the content is restricted from both

transversal and longitudinal movements within between the at least two sleeves, by an interlocking retention between a couple of interlocking components respectively formed on the at least two sleeves.

Preferably, the covering body and the base seat both are substantially the same in their entire structure and dimension, wherein while the covering body, the sleeving unit and the base seat are assembled together into the entire stick-shaped paper container, the covering body and the base seat both are disposed in a reverse-position manner with relative to each other.

Preferably, after the covering body and the sleeving unit both are assembled together, the covering body and the sleeving unit both are respectively positioned with the corresponding positive draft angles relative to the longitudinal centre line.

Preferably, after the covering body, the sleeving unit and the base seat are assembled together into the entire stick-shaped paper container, the second end of the sleeving unit rests on an inner sidewall of the bottom terminal of the base seat.

Preferably, the outermost cover sidewall and the outermost base sidewall both are respectively formed with flatted surfaces extended toward the same direction.

Preferably, a transversally cross-sectional thickness of the standing cover wall is smaller or larger than a transversally cross-sectional thickness of the standing base wall.

Preferably, the innermost seat sidewall is divided, by along the longitudinal centre line, into an annularly-narrowed inner diameter section, an annularly-widened inner diameter section, and an inner circular saddle formed on a boundary between both the annularly-narrowed inner diameter section and the annularly-widened inner diameter, wherein the second end passes through the receiving opening to enter inside the hollowed base chamber while the abutting terminal is disposed with the second end, the annularly-widened inner diameter section and the outermost sleeve sidewall both are respectively formed with flatted surfaces extended toward the same direction in such a way of constituting a plane-to-plane tight contact in a mutually resting-on manner between the annularly-widened inner diameter section and the outermost sleeve sidewall, and an outer diameter of the second end is smaller than an inner diameter of the inner circular saddle, thereby blocking the second end on the inner circular saddle from moving.

Preferably, an outer diameter of the abutting terminal of the base seat is smaller than an inner diameter of the second opening of the sleeving unit and smaller than an inner diameter of the fitting-on opening of the covering body.

Preferably, while the container height-to-width ratio is between 2~5.5, the maximum container width is between 12~23 mm.

Furthermore, another preferred embodiment of the present invention adopts the following technical solutions where a stick-shaped paper container, configured for accommodating a content therein, comprises a covering body and a base seat.

The covering body has a closed top distal end, a fitting-on distal end opposed against the top distal end, a standing cover wall located between the top distal end and the fitting-on distal end and formed around the longitudinal centre line, and a hollowed cover chamber formed between the top distal end and the fitting-on distal end, wherein the fitting-on distal end is formed with a fitting-on opening space-communicated with the hollowed cover chamber, and the hollowed cover chamber is formed with an innermost cover sidewall therein.

The base seat is configured for longitudinally supporting the content and has a closed bottom terminal, an abutting terminal opposed against the bottom terminal, a hollowed base chamber formed between the bottom terminal and the abutting terminal, and a standing base wall located between the bottom terminal and the abutting terminal and formed around the longitudinal centre line, wherein the abutting terminal is formed with a receiving opening space-communicated with the hollowed base chamber, the abutting terminal of the base seat is disposed at a position corresponding to the fitting-on distal end of the covering body, thereby accommodating the content within between both the covering body and the base seat.

Preferably, the covering body and the base seat both are integrally formed, respectively, only by a wet-fiber pulp-molding process where a male mold dredges up wet plant-fibrous pulps and then the male mold and a female mold both compression-mold the wet plant-fibrous pulps, in such a way that the entire structure of each of the covering body and the base seat all is formed with positive draft angles relative to the longitudinal centre line, and the stick-shaped paper container is constituted only by assembling the covering body and the base seat together.

Preferably, the stick-shaped paper container has a maximum container height parallel to the longitudinal centre line, a maximum container width perpendicular to the longitudinal centre line, and a container height-to-width ratio, between 2~5.5, of the maximum container height being relative to the maximum container width.

Furthermore, another preferred embodiment of the present invention adopts the following technical solutions where a stick-shaped paper container, configured for accommodating a content therein, comprises a sleeving unit and a covering body.

The sleeving unit configured for restricting the content from movement has a first and second end both opposed against each other, an annularly standing sleeve wall located between the first and second ends and formed around a longitudinal centre line, and a hollowed sleeve chamber that the longitudinal centre line passes through being formed between the first and second ends, wherein the first end is formed with a first opening space-communicated with the hollowed sleeve chamber and provided for the content passing therethrough, the hollowed sleeve chamber is formed with an innermost sleeve sidewall in a position of propping the content, and the standing sleeve wall is formed with an outermost sleeve sidewall.

The covering body has a closed top distal end, a fitting-on distal end opposed against the top distal end, a standing cover wall located between the top distal end and the fitting-on distal end and formed around the longitudinal centre line, and a hollowed cover chamber formed between the top distal end and the fitting-on distal end, wherein the fitting-on distal end is formed with a fitting-on opening space-communicated with the hollowed cover chamber, the hollowed cover chamber is formed with an innermost cover sidewall therein, the covering body is disposed at a position corresponding to the sleeving unit.

Preferably, after the covering body and the sleeving unit both are integrally formed, respectively, only by a wet-fiber pulp-molding process where a male mold dredges up wet plant-fibrous pulps and then the male mold and a female mold both compression-mold the wet plant-fibrous pulps, the entire structure of each of the covering body and the sleeving unit is formed with positive draft angles relative to the longitudinal centre line, and the stick-shaped paper

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container is constituted only by assembling the covering body and the sleeving unit together.

Preferably, the innermost cover sidewall is divided, by along the longitudinal centre line, into an annularly-narrowed inner diameter section, an annularly-widened inner diameter section, and a first outer circular saddle formed on an end of the annularly-widened inner diameter section and outwardly extended, and the outermost sleeve sidewall is outwardly extended and is formed with a second outer circular saddle thereon, wherein the first end of the sleeving unit passes through the fitting-on opening to enter inside the hollowed cover chamber while the covering body is disposed with the sleeving unit, the annularly-widened inner diameter section and the outermost sleeve sidewall both are respectively formed with smoothly curved surfaces obliquely extended in such a way of constituting a plane-to-plane tight contact in a mutually resting-on manner between the annularly-widened inner diameter section and the outermost sleeve sidewall, and an outer diameter of the second end is larger than an inner diameter of the fitting-on opening, thereby blocking the second outer circular saddle on a bottom side of the first outer circular saddle from moving.

Consequently, the present invention can effect the following technical benefits that: compared with the prior arts, each of all components of the stick-shaped paper container according to the present invention, is capable of being integrally molded, respectively, by consistent-and-continuous production machines of a production line used with an improved wet-fibrous pulp-molding process. This does not only resolve the technical drawback of the conventional pulp-molding process that is incapable of producing such a pulp-molded article having a height-to-width ratio, greater than one, of its maximum longitudinal height being relative to maximum transversal width, but also reduces its working cycle time, benefits its mass production, and assures its higher product yield and quality. Furthermore, each of all components of the stick-shaped paper container according to the present invention, has the respective demold surfaces all rendered only in positive draft angles relative to a longitudinally demolding direction of a compression-molding mold assembly used with the wet-fiber pulp-molding process, rather than negative draft angles relative to the longitudinally demolding direction, as rendered in the laterally latch structure and/or laterally recessed structure of the conventional pulp-molded article. Thus, the present invention does not need to additionally use the other transversal mold implemented for a horizontally demolding, and/or a slidable block device implemented for a horizontal movement, thereby being capable of simplifying the mold assembly, and lowering its mold cost.

DESCRIPTION OF THE DIAGRAMS

FIG. 1 depicts a schematic diagram of consistent production machines allocated in a production line used with a conventional pulp-molding process;

FIG. 2 depicts a schematically cross-sectional diagram of consistently automatic production machines allocated in a production line used with an improved wet-fiber pulp-molding process, according to the present invention;

FIG. 3A depicts a perspective diagram of a stick-shaped paper container according to the first preferred embodiment of the present invention;

FIG. 3B depicts an enlarged laterally cross-sectional view according to the stick-shaped paper container shown in FIG. 3A;

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FIG. 3C depicts an exploded view according to the stick-shaped paper container shown in FIG. 3A;

FIG. 4A depicts a perspective diagram of a stick-shaped paper container according to a first preferred embodiment of the present invention;

FIG. 4B depicts an enlarged cross-sectional view according to the stick-shaped paper container shown in FIG. 4A;

FIG. 4C depicts an exploded view according to the stick-shaped paper container shown in FIG. 4A;

FIG. 5A depicts a perspective diagram of a stick-shaped paper container according to a third preferred embodiment of the present invention;

FIG. 5B depicts an enlarged cross-sectional view according to the stick-shaped paper container shown FIG. 5A;

FIG. 5C depicts an exploded view according to the stick-shaped paper container shown FIG. 5A;

FIG. 6A depicts a perspective diagram of a stick-shaped paper container according to a fourth preferred embodiment of the present invention;

FIG. 6B depicts an enlarged cross-sectional view according to the stick-shaped paper container shown in FIG. 6A;

FIG. 6C depicts an exploded view according to the stick-shaped paper container shown in FIG. 6A;

FIG. 7A depicts a perspective diagram of a stick-shaped paper container 30 according to a fifth preferred embodiment of the present invention;

FIG. 7B depicts an enlarged cross-sectional view according to the stick-shaped paper container shown in FIG. 7A;

FIG. 7C depicts an exploded view according to the stick-shaped paper container shown in FIG. 7A; and

FIG. 7D depicts a partially assembled view according to the stick-shaped paper container shown in FIG. 7A.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The technical proposals in the embodiments of the present invention will be clearly and completely described in the following with reference to the accompanying drawings of the embodiments of the present invention. The directional terms mentioned in the present invention, such as "upper", "lower", "before", "after", "left", "right", "inside", "outside", "side", etc., are merely illustrative the direction of the drawing. Therefore, the directional terminology used is for the purpose of illustration and understanding of the invention, which is not intended to limit the invention.

First of all, please refer to an illustration shown in FIG. 2, which depicts a schematically cross-sectional diagram of consistently automatic production machines 70 allocated in a production line used with an improved wet-fiber pulp-molding process, according to the present invention. The consistently automatic production machines 70 allocated in the production line used with the improved wet-fiber pulp-molding process, is primarily configured with at least one actuating-to-move apparatus (not shown), a dredging-pulp and pre-compression device 80, a vacuum exhausting device (not shown), a thermo-compression-molding device 90 and a trimming-off device 98.

The dredging-pulp and pre-compression device 80 comprises a first female mold 82 located on an upper portion thereof, and a first male mold 84 located on a lower portion thereof with corresponding to the first female mold 82. In this present invention, the herein-called 'female mold' is defined as such a mold that has an outer molding surface which a caved structure is caved inwardly from, and is primarily operable to shape outer circumferential surfaces of each of all components of a paper article such as a stick-

shaped paper container, and the herein-called 'male mold' is defined as such a mold that has an outer molding surface which a protrusive structure is protruded outwardly from, and is primarily operable to shape inner circumferential surfaces of all components of a paper article as a stick-shaped paper container.

As shown in FIG. 2, the first male mold **84** is disposed with a plurality of spaced-apart first upright posts **85** thereon, which are allocated in a multidimensional-array manner with even distribution, and are protruded outwardly, along a vertical direction, from an upper surface of the first male mold **84**. Each of the first upright posts **85** of the first male mold **84** is formed with a longitudinal outermost wall surface substantially rendered as a cylindrical surface, a conical-frustum surface, a truncated-cone surface or a bell-shaped curve surface, but its shape is not therefore limited thereto. The plurality of first upright posts **85** are capable to massively producing the respective components of a stick-shaped paper container **10** (See FIG. 3A) at the same cycle time. In the present invention, each of the first upright posts **85** has a maximum first-upright-post height **H2** formed parallel to a corresponding longitudinally mold-matching centre line **Y1** but perpendicular to a first-upright-post height **H2**, wherein a post height-to-width ratio **R2** that the maximum first-upright-post height **H2** is relative to the maximum first-upright-post width **W2** is greater than one (i.e. $H2/W2=R2$, $R2>1$). Nevertheless, in another preferred embodiment for the demand on practical production, it can be replaced with other first upright posts **85** having a post height-to-width ratio **R2** smaller than one.

Correspondingly, the first female mold **82** as shown in FIG. 2, has a plurality of spaced-apart first vertical pits **822** formed inside a bottom surface of the first female mold **82**, with the same spaced intervals thereamong as same as allocated among the plurality of first upright posts **85**, such that the plurality of first vertical pits **822** have a deployed arrangement and a sized proportion both respectively corresponding to and being aligned with a deployed arrangement and a sized proportion of the plurality of first upright posts **85**. Each of the first vertical pits **822** is formed with a longitudinally inner circumferential wall surface can also be substantially rendered as a cylindrical surface, a conical-frustum surface, a truncated-cone surface or a bell-shaped curve surface.

Furthermore, the entire outermost wall surface of the respective first upright posts **85** of the first male mold **84** and the entire inner circumferential wall surface of the first vertical pits **822** of the first female mold **82** both are respectively formed thereon with a number of micro-pores (not shown) in an evenly-distributing manner. The number of micro-pores are respectively individually liquid-communicated with a vacuum exhausting device, via number of exhausting passages respectively defined inside both the first male mold **84** and the first female mold **82** so as to exhaust moistures and/or their existed inside a pulp (only by way of vacuum exhausting) which is located over the longitudinal outermost wall surface of the respective first upright posts **85**.

As depicted in FIG. 2, during a production process, such as the improved wet-fiber pulp-molding process, of the consistently automatic production machines **70** allocated in the production line, at an initial stage the dredging-pulp and pre-compression device **80** makes the first male mold **84** sunk into a slurry tank **86** for storing a slurry (or called 'pulp') **41** that contains a large amount of wet plant fibrous pulps **81**; and next, only by way of vacuum exhausting of the vacuum exhausting device via the exhausting passages of

the first male mold **84**, the wet plant-fibrous pulps **81** are evenly adsorbed into a layer over the entire longitudinal outermost wall surface of the respective first upright post **8** of the first male mold **84**. Next, the dredging-pulp and pre-compression device **80** makes the first female mold **82** and the first male mold **84** both being respectively upwardly-and-downwardly moved to be mutually matched with each other and further exerts a slight pressure to pre-compress the wet plant-fibrous pulps **81** located between the first female mold **82** and the first male mold **84**, wherein the respective first vertical pits **822** of the first female mold **82** and the corresponding first upright posts **85** of the first male mold **84** both are mutually matched, commonly along the corresponding longitudinally mold-matching centre line **Y1**, thereby integrally compression-molding the respective component of the stick-shaped paper container **10** (See FIG. 3A); simultaneously, by the vacuum exhausting device implementing the afore-mentioned vacuum exhausting between the first male and female molds **84**, **82**, a vacuum environment is established therebetween by exhausting out a less portion of water vapor and/or moisture contained in the wet plant-fibrous pulps **81**, so as to integrally form a wet paper article **42** (or called 'wet billet'), constructed of the wet plant fibrous pulps **81**, between the first female mold **82** and the first male mold **84**. In a practical exemplar, a material ingredient of the pulp **81** includes a composition of bamboo pulps and bagasse pulps, but is not limited thereto such a composition ingredient of the wet pulps **81**, the working pressure range and a working temperature range of the pulp-dredging and pre-compression device **80**, and a moisture content range of the wet paper article **42**. This is because depending on different product structures and demands, the composition ingredient and the proportion of the wet pulp **81**, the working pressure range and the working temperature range used in the pulp-dredging and pre-compression device **80**, and the moisture content range of the wet paper article **42** all might be changed.

Next, as illustrated in FIG. 2, by the vacuum exhausting device vacuum-suctioning the wet paper article **42** onto an underside of the bottom surface of the first female mold **82**, the at least one actuating-to-move apparatus makes the first female mold **82** moved with bringing the adsorbed wet paper article **42** together, to reach between both a second female mold **92** and a second male mold **94** of thermo-compression-molding device **90**.

Next, by relieving the vacuum suction, the first female mold **82** releases the wet paper article **42** to be positioned over the second male mold **94** of the thermo-compression-molding device **90**. The second female mold **92** and the second male mold **94** of the thermo-compression-molding device **90** have a deployed arrangement and a sized proportion (including size proportions of each of the coupled molds **92**, **94**) which are similar to a deployed arrangement and a sized proportion of both the first female mold **82** and the first male mold **84** of the dredging-pulp and pre-compression device **80**. For example, an upper surface of the second male mold **94** is disposed with a plurality of spaced-apart second upright posts **942** in the same deployed arrangement and the same sized proportion as used in forming the plurality of first upright posts **85** on the first male mold **84**, and a plurality of spaced-apart second vertical pits **922** are inwardly formed inside a bottom surface of the second female mold **92**, in the same deployed arrangement and the same sized proportion as used in forming the plurality of first vertical pits **822** on the first female mold **82**.

Next, as illustrated in FIG. 2, the thermo-compression-molding device **90** makes the second female mold **92** and the

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second male mold **94** both being respectively upwardly-and-downwardly moved to be mutually matched with each other, and exerts a higher pressure to thermally compress the wet paper article **42** positioned between the second female mold **92** and the second male mold **94**; simultaneously, by the way of the vacuum exhausting of the vacuum exhausting device used with both the second female mold **92** and the second male mold **94** of the thermo-compression-molding device **90**, a larger portion of water vapor and/or moisture contained in the wet paper article **42** located between the second female mold **92** and the second male mold **94** is exhausted out so as to form a dried paper article **44** constructed of the dried plant fibrous pulps **81**.

Next, as illustrated in FIG. 2, the trimming-off device **98** is operable to cut away a few superfluous portions from the dried paper article **44**, thereby respectively forming the respective components of the stick-shaped paper container **10** (as a covering body **12** shown in FIG. 3A, which will be detailed later), which is constructed of absolutely-dried plant fibrous pulps **81**. In this embodiment, the trimming-off device **98** might be an existing duplicating-to-cut circumferential cutting machine having cutting molds, or any other kinds of trimming-off device.

Please further refer to FIGS. 2 & 3A-3C, wherein FIG. 3A depicts a perspective diagram of the stick-shaped paper container **10** according to a first preferred embodiment of the present invention. The stick-shaped paper container **10** is configured to encapsulation-accommodate a content **100** therein, and primarily comprises the following components: a covering body **12**, a sleeving unit **14** and a base seat **16**. In this preferred embodiment, the covering body **12**, the sleeving unit **14** and the base seat **16** all are integrally molded, respectively, only by preparations of the consistently automatic production machines **70** allocated in the production line used with the improved wet-fiber pulp-molding process, as depicted in FIG. 2. The herein-called 'content' **100** is a main body having a solid-state structure, including but being not limited to, for example cosmetics, lipsticks, 3C electronic products, ornaments, or any thinned-and-elongated appliances/composition for specific function and so forth.

Further referring to FIGS. 3A-3C, the sleeving unit **14** is a single sleeve for tightly sheathing over an outermost wall surface of the content **100** and thereby restricting the content **100** from transversal movement. The sleeving unit **14** is formed with two opposed ends **142**, **144** (such as a first end **142** and a second end **144**), an annularly standing sleeve wall **146** located between the first and second ends **142**, **144** and surrounding a longitudinal centre line G1 (parallel to the longitudinally mold-matching centre line Y1 shown in FIG. 2), and a hollowed sleeve chamber **148** which the longitudinal centre line G1 passes through and which is formed between the first and second ends **142**, **144**. The first end **142** is formed with a first opening **1422** thereon space-communicated with the hollowed sleeve chamber **148** and provided for the content **100** upwardly passing therethrough. The second end **144** is formed with a second opening **1442** thereon space-communicated with the hollowed sleeve chamber **148** and provided for the content **100** downwardly passing therethrough. The hollowed sleeve chamber **148** configured for accommodating the content **100** therein is formed with an innermost sleeve sidewall **1482** at a position of propping the content **100**. The standing sleeve wall **146** has an outermost sleeve sidewall **1462** that is integrally compression-molded, with dependence on a contour of the longitudinally inner circumferential wall surface of the corresponding respective first vertical pit **822** (as shown in FIG. 2), into an outer cylindrical surface or an outer truncated-cone surface but its shape is not therefore limited thereto. Similarly, the innermost sleeve sidewall **1482** of the hollowed sleeve chamber **148** is integrally compression-molded, with dependence on a contour of the longitudinally outermost wall surface of the corresponding respective first upright post **85** (as shown in FIG. 2), into an inner cylindrical surface or an inner truncated-cone surface but its shape is not therefore limited thereto.

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Further referring to FIGS. 3A-3C, the outermost sleeve sidewall **1462** is downwardly sequentially divided, from the first end **142** along the longitudinal centre line G1, into a gradually-widened multi-stage portion having different inner diameter per sectioned stage, which comprises an annularly outer narrow-diameter section **1464**, an annularly outer wide-diameter section **1466**, and a first outer circular saddle **1468** formed on a boundary between the outer narrow-diameter section **1464** and the outer wide-diameter section **1466**. An outer diameter of the outer wide-diameter section **1466** is larger than an outer diameter of the outer narrow-diameter section **1464** but substantially equal to an outer diameter of the outer circular saddle **1468**. The innermost sleeve sidewall **1482** of the hollowed sleeve chamber **148** is configured to prop up the content **100** and is downwardly divided, from the second end **144** along the longitudinal centre line G1, into a gradually-narrowed multi-stage portion having different inner diameter per sectioned stage, which comprises an annularly-widened inner diameter section **1484**, a first annularly-narrowed inner diameter section **1486**, a first inner circular saddle **1488** formed on a boundary between the annularly-widened inner diameter section **1484** and the first annularly-narrowed inner diameter section **1486**, a second annularly-narrowed inner diameter section **1489**, and a second inner circular saddle **1490** formed on a boundary between the first annularly-narrowed inner diameter section **1486** and the second annularly-narrowed inner diameter section **1489**. An inner diameter of the annularly-widened inner diameter section **1484** is larger than an inner diameter of each of the first annularly-narrowed inner diameter section **1486**, the second annularly-narrowed inner diameter section **1489**, the first inner circular saddle **1488**, and the second inner circular saddle **1490**, as well as the inner diameter of the first annularly-narrowed inner diameter section **1486** is larger than the inner diameter of the second annularly-narrowed inner diameter section **1489** and the inner diameter of the first inner circular saddle **1488** is larger than the inner diameter of the second inner circular saddle **1490**, whereby the second inner circular saddle **1490** and the second annularly-narrowed inner diameter section **1489** both are capable of just transversally supporting and tightly restricting the content **100** on the base seat **16**.

Further referring to FIGS. 3A-3C, the covering body **12** has a closed top distal end **122**, a fitting-on distal end **124** opposed against the top distal end **122**, a standing cover wall **126** formed between the top distal end **122** and the fitting-on distal end **124** and around the longitudinal centre line G1, and a hollowed cover chamber **128** formed between the top distal end **122** and the fitting-on distal end **124**. The fitting-on distal end **124** is formed with a fitting-on opening **1242** space-communicated with the hollowed cover chamber **128** and is disposed at a position corresponding to the first end **142** of the sleeving unit **14**. The hollowed cover chamber **128** is formed with an innermost cover sidewall **1282** therein for propping up the content **100**. The standing cover wall **126** has an outermost cover sidewall **1262**. Based on the same fabricating principle, the outermost cover sidewall **1262** of the standing cover wall **126** is integrally compression-molded, with dependence on a contour of the longitudinally inner circumferential wall surface of the corresponding respective first vertical pit **822** (as shown in FIG. 2), into an outer cylindrical surface or an outer truncated-cone surface but its shape is not therefore limited thereto. Similarly, the innermost cover sidewall **1282** of the hollowed cover chamber **128** is integrally compression-molded, with dependence on a contour of the longitudinally outermost wall surface of the corresponding respective first upright post **85** (as shown in FIG. 2), into an inner cylindrical surface or an inner truncated-cone surface but its shape is not therefore limited thereto.

Further referring to FIGS. 3A-3C, the covering body **12** has a closed top distal end **122**, a fitting-on distal end **124** opposed against the top distal end **122**, a standing cover wall **126** formed between the top distal end **122** and the fitting-on distal end **124** and around the longitudinal centre line G1, and a hollowed cover chamber **128** formed between the top distal end **122** and the fitting-on distal end **124**. The fitting-on distal end **124** is formed with a fitting-on opening **1242** space-communicated with the hollowed cover chamber **128** and is disposed at a position corresponding to the first end **142** of the sleeving unit **14**. The hollowed cover chamber **128** is formed with an innermost cover sidewall **1282** therein for propping up the content **100**. The standing cover wall **126** has an outermost cover sidewall **1262**. Based on the same fabricating principle, the outermost cover sidewall **1262** of the standing cover wall **126** is integrally compression-molded, with dependence on a contour of the longitudinally inner circumferential wall surface of the corresponding respective first vertical pit **822** (as shown in FIG. 2), into an outer cylindrical surface or an outer truncated-cone surface but its shape is not therefore limited thereto. Similarly, the innermost cover sidewall **1282** of the hollowed cover chamber **128** is integrally compression-molded, with dependence on a contour of the longitudinally outermost wall surface of the corresponding respective first upright post **85** (as shown in FIG. 2), into an inner cylindrical surface or an inner truncated-cone surface but its shape is not therefore limited thereto.

dinally inner circumferential wall surface of the corresponding respective first vertical pits **822** (as shown in FIG. 2), into an outer cylindrical surface or an outer truncated-cone surface, but its shape is not therefore limited thereto. Similarly, the innermost cover sidewall **1282** of the hollowed cover chamber **128** is integrally compression-molded, with dependence on a contour of the longitudinal outermost wall surface of the corresponding respective first upright post **85** (as shown in FIG. 2), into an inner cylindrical surface or an inner truncated-cone surface, but its shape is not therefore limited thereto.

Further referring to FIGS. 3A-3C, the base seat **16** is configured for longitudinally supporting a bottom side of the content **100**, and has a closed bottom terminal **162**, an abutting terminal **164** opposed against the bottom terminal **162**, a hollowed base chamber **166** formed between the bottom terminal **162** and the abutting terminal **164**, and a standing base wall **168** formed between the bottom terminal **162** and the abutting terminal **164** and around the longitudinal centre line G1. The abutting terminal **164** is formed with a receiving opening **1642** space-communicated with the hollowed base chamber **166** and is disposed at a position corresponding to the second end **144** of the sleeving unit **14**, thereby being capable of encapsulation-accommodating the content **100** in between the covering body **12** and the base seat **16**. The standing base wall **168** has an outermost base sidewall **1682**. Based on the same fabricating principle, the outermost base sidewall **1682** of the standing base wall **168** is integrally compression-molded, with dependence on a contour of the longitudinally inner circumferential wall surface of the corresponding respective first vertical pit **822** (as shown in FIG. 2), into an outer cylindrical surface or an outer truncated-cone surface, but its shape is not therefore limited thereto. Similarly, the innermost seat sidewall **1684** of the hollowed base chamber **166** is integrally compression-molded, with dependence on a contour of the longitudinal outermost wall surface of the corresponding respective first upright post **85** (as shown in FIG. 2), into an inner cylindrical surface or an inner truncated-cone surface, but its shape is not therefore limited thereto.

Further referring to FIGS. 2 & 3A-3C which depict the first preferred embodiment, the covering body **12**, the sleeving unit **14** and the base seat **16** all are individually fabricated only by the consistently automatic production machines **70** (as depicted in FIG. 2), allocated in the production line used with the improved wet-fiber pulp-molding process, where the first male mold **84** is used to dredge up the wet plant-fibrous pulps **81** and then the first male mold **84** and the first female mold **82** both are mutually matched for compression-molding from the wet plant-fibrous pulps **81** into an integral. This leads the respective demold surfaces of the entire structure of each of the covering body **12**, the sleeving unit **14** and the base seat **16** to be formed with positive draft angles $\theta 1$ (i.e. smaller than or equal to 1.5 degrees) relative to the longitudinal centre line G1 (parallel to the longitudinally mold-matching centre line Y1 depicted in FIG. 2), and to make the outermost cover sidewall **1262**, the outermost sleeve sidewall **1462** and the outermost base sidewall **1682** all being formed respectively with flatted surfaces extended toward the same direction.

Further referring to FIGS. 3A-3C, while the fitting-on distal end **124** of the covering body **12** is disposed with the first end **142** of the sleeving unit **14**, the first end **142** passes through the fitting-on opening **1242** to enter inside the hollowed cover chamber **128**. The innermost cover sidewall **1282** and the outer narrow-diameter section **1464** both are respectively formed with flatted surfaces longitudinally

extended in such a way of constituting a plane-to-plane tight contact in a mutually-and-longitudinally resting-on manner (such as a loose fit) between the innermost cover sidewall **1282** and the outer narrow-diameter section **1464**. An inner diameter of the fitting-on opening **1242** is equal to the outer diameter of the outer circular saddle **1468** but is smaller than an outer diameter of the outer wide-diameter section **1466**, thereby blocking the fitting-on distal end **124** on the outer circular saddle **1468** from moving, and blocking the outer wide-diameter section **1466** of the sleeving unit **14** from continuously entering inside the hollowed cover chamber **128**. Preferably, after the covering body **12** and the sleeving unit **14** both are assembled together, the covering body **12** and the sleeving unit **14** both still are respectively rendered in the corresponding positive draft angles $\theta 1$ (See FIG. 3A) if the longitudinal centre line G1 is treated as a datum line relative to each position.

Further referring to FIGS. 3A-3C, while the abutting terminal **164** of the base seat **16** is disposed with the second end **144** of the sleeving unit **14**, the abutting terminal **164** passes through the second opening **1442** of the sleeving unit **14** to enter inside the hollowed sleeve chamber **148**. The annularly-widened inner diameter **1484** and the outermost base sidewall **1682** both are respectively formed with flatted surfaces longitudinally extended in such a way of constituting a plane-to-plane tight contact in a mutually-and-longitudinally resting-on manner between the annularly-widened inner diameter **1484** and the outermost base sidewall **1682** (or adding other permanently retaining structure such as a glue into therebetween). An outer diameter of the abutting terminal **164** is equal to the inner diameter of the first inner circular saddle **1488** but is smaller than the inner diameters of both the first annularly-narrowed inner diameter section **1486** and the fitting-on opening **124** of the covering body **12**, thereby blocking the abutting terminal **164** on the first inner circular saddle **1488** from moving to continuously enter inside the first annularly-narrowed inner diameter section **1486**. Preferably, the outer diameter of the abutting terminal **164** of the base seat **16** is smaller than the inner diameters of both the second opening **1442** of the sleeving unit **14** and the fitting-on opening **1242** of the covering body **12**.

Further referring to FIGS. 3A-3C, after the covering body **12**, the sleeving unit **14** and the base seat **16** all are assembled together into the entire stick-shaped paper container **10**, the stick-shaped paper container **10** has a maximum container height $h2$ parallel to the longitudinal centre line G1, a maximum container width $w2$ perpendicular to the longitudinal centre line G1, and a container height-to-width ratio $r2$, between $2 \sim 5.5$ (i.e. $h2/w2 = r2$, $5.5 \geq r2 \geq 2$), of the maximum container height $h2$ being relative to the maximum container width $w2$. Preferably, while the container height-to-width ratio $r2$ is between $2 \sim 5.5$, the maximum container width $w2$ is between 12~23 mm. Preferably, after the covering body **12**, the sleeving unit **14** and the base seat **16** all are assembled together, the base seat **16** is deployed in a reverse-position manner relative to both the covering body **12** and the sleeving unit **14** if the longitudinal centre line G1 is treated as a datum line relative to each position.

Furthermore, please refer to the illustrations shown in FIGS. 2 & 4A-4C wherein FIG. 4A depicts a perspective diagram of a stick-shaped paper container **20** according to a second preferred embodiment of the present invention. In the second preferred embodiment, the stick-shaped paper container **20** has the similar assembled components, as all above mentioned, comprising a covering body **12**, a sleeving unit **14** and a base seat **16**. However, the detailed structures and deployed arrangements of each of the covering body **12**,

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the sleeving unit **14** and the base seat **16** in the stick-shaped paper container **20** of the second preferred embodiment shown in FIGS. 4A-4C have the following differences, from the stick-shaped paper container **10** of the first preferred embodiment shown in FIGS. 3A-3C, that:

- (1) In the second preferred embodiment depicted in FIGS. 4A-4C, both the covering body **12** and the sleeving unit **14** of the stick-shaped paper container **20** are respectively fabricated, only by the consistently automatic production machines **70** allocated in the production line used with the improved wet-fiber pulp-molding process as depicted in FIG. 2, where the first male mold **84** is used to dredge up the wet plant-fibrous pulps **81** and then the first male mold **84** and the first female mold **82** both are mutually matched for compression-molding from the wet plant-fibrous pulps **81** into an integral. Nevertheless, the base seat **16** can be replaced with using of a typical cylindrical paper tube for lowering its fabricated cost, but is not therefore limited thereto. In another preferred embodiment, the covering body **12**, the sleeving unit **14** and the base seat **16** still can be integrally compression-molded, respectively, only by the consistently automatic production machines **70** allocated in the production line used with the improved wet-fiber pulp-molding process, as depicted in FIG. 2;
- (2) After integrally compression-molding by the consistently automatic production machines **70** allocated in the production line used with the improved wet-fiber pulp-molding process (as depicted in FIG. 2), the corresponding demold surfaces of the entire structure of each of the covering body **12** and the base seat **16** are formed with positive draft angles $\theta 1$ (i.e. smaller than or equal to 1.5 degrees) relative to the longitudinal centre line G1 (parallel to the longitudinally mold-matching centre line Y1 depicted in FIG. 2), thereby making both the outermost cover sidewall **1262** of the covering body **12** and the outermost base sidewall **1682** of the base seat **16** being respectively formed with flatted surfaces extended toward the same direction; nevertheless, after the covering body **12**, the sleeving unit **14** and the base seat **16** all are assembled together into the entire stick-shaped paper container **20** as shown in FIG. 4A, the covering body **12** and the base seat **16** both are deployed, in a reverse-position manner relative to each other, as two truncated-cones stacked from up to down.
- (3) Further referring to FIGS. 4A-4C which depict the second preferred embodiment, while the abutting terminal **164** of the base seat **16** is disposed with the second end **144** of the sleeving unit **14** (as a closed end for longitudinally supporting the content **100**), the second end **144** of the sleeving unit **14** can enter inside the hollowed base chamber **166** of the base seat **16** until the second end **144** of the sleeving unit **14** rests on an inner sidewall of the bottom terminal **162** of the base seat **16** since an inner diameter of the receiving opening **1642** of the abutting terminal **164** of the base seat **16** is larger than an outer diameter of the outermost sleeve sidewall **1462** of the sleeving unit **14**. The outermost sleeve sidewall **1462** of the sleeving unit **14** and the innermost seat sidewall **1684** of the base seat **16** both are respectively formed with flatted surfaces longitudinally extended in such a way of constituting a plane-to-plane tight contact in a mutually-and-longitudinally resting-on manner between the outermost sleeve sidewall **1462** and the innermost seat sidewall **1684** (or adding other permanently retaining structure such as a

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glue into therebetween); and since a height of the standing base wall **168** is smaller than a height of the standing sleeve wall **146**, thereby directing the first end **142** of the sleeving unit **14** to position beyond and outside the abutting terminal **164** of the base seat **16**;

- (4) Further referring to FIGS. 4A-4C which depict the second preferred embodiment, the first end **142** passes through the fitting-on opening **1242** of the covering body **12** to enter inside the hollowed cover chamber **128** while the fitting-on distal end **124** is disposed with the first end **142** of the sleeving unit **14**. The innermost cover sidewall **1282** and the outermost sleeve sidewall **1462** both are respectively formed with flatted surfaces longitudinally extended in such a way of constituting a plane-to-plane tight contact (as a loose fit) in a mutually-and-longitudinally resting-on manner between the innermost cover sidewall **1282** and the outermost sleeve sidewall **1462**; and an inner diameter of the fitting-on opening **1242** is larger than an outer diameter of the first end **142** but an outer diameter the fitting-on distal end **124** of the covering body **12** is substantially equal to an outer diameter of the abutting terminal **164** of the base seat **16**, thereby making the fitting-on distal end **124** of the covering body **12** resting on and being blocked on the abutting terminal **164** of the base seat **16**; and
- (5) Further referring to FIG. 4B, a transversally cross-sectional thickness of the standing cover wall **126** of the covering body **12** is larger than a transversally cross-sectional thickness of each of both the standing base wall **168** and the standing sleeve wall **146**, but is not therefore limited thereto. For example, in another preferred embodiment, the covering body **12** and the base seat **16** both can be substantially the same in their entire structures and dimensions.

Except for the afore-mentioned different structures, the other remaining structures of each of the covering body **12**, the sleeving unit **14** and the base seat **16** of the stick-shaped paper container **20** according to the second preferred embodiment shown in FIGS. 4A-4C all are substantially the same as the stick-shaped paper container **10** according to the first preferred embodiment shown in FIGS. 3A-3C in both preparations and allocations thereof, and therefore all can be referred to the afore-mentioned introduction for the respective components of the stick-shaped paper container **10** of the first preferred embodiment shown in FIGS. 3A-3C, but will be omitted in detail later. For example, as shown in FIG. 4B, after the covering body **12**, the sleeving unit **14** and the base seat **16** all are assembled together into the entire stick-shaped paper container **20**, the stick-shaped paper container **20** has a maximum container height $h2$ parallel to the longitudinal centre line G1, a maximum container width $w2$ perpendicular to the longitudinal centre line G1 and a container height-to-width ratio $r2$ of the maximum container height $h2$ being relative to the maximum container width $w2$; consistently, the container height-to-width ratio $r2$ is between $2 \sim 5.5$ (namely $h2/w2=r2$, $5.5 \geq r2 \geq 2$). Preferably, while the container height-to-width ratio $r2$ is between $2 \sim 5.5$, the maximum container width $w2$ is between $12 \sim 23$ mm.

Nevertheless, it should be noted that in another embodiment based on an improvement to the second preferred embodiment, the stick-shaped paper container **20** could be assembled with only using both the covering body **12** and the base seat **16** but without the need of using the sleeving unit **14**, wherein a lower portion of the content **100** could be permanently retained within the base seat **16** as well as an

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upper portion of the content **100** is sheathed in an encapsulation manner by the hollowed cover chamber **128** of the covering body **12**, thereby constituting a plane-to-plane loose fit in a manner that the innermost cover sidewall **1282** of the covering body **12** directly props up the upper portion of the content **100**.

Furthermore, please refer to the illustrations shown in FIGS. **5A-5C**, wherein FIG. **5A** depicts a perspective diagram of the stick-shaped paper container **30** according to a third preferred embodiment of the present invention. In the third preferred embodiment, the stick-shaped paper container **30** has the similar assembled components, as all above mentioned, comprising a covering body **12**, a sleeving unit **14** and a base seat **16**. Nevertheless, the detailed structure and deployed arrangement of each of the covering body **12**, the sleeving unit **14** and the base seat **16** in the stick-shaped paper container **30** according to the third preferred embodiment shown in FIGS. **5A-5C** have the following differences, from the stick-shaped paper container **20** according to the second preferred embodiment shown in FIGS. **4A-4C**, that:

- (1) After integrally compression-molding by the consistently automatic production machines **70** allocated in the production line used with the improved wet-fiber pulp-molding process (as depicted in FIG. **2**), the corresponding demold surfaces of the respective entire structure of both the covering body **12** and the base seat **16** are formed with positive draft angles $\theta 1$ (i.e. smaller than or equal to 1.5 degrees) relative to the longitudinal centre line **G1** (parallel to the longitudinally mold-matching centre line **Y1** depicted in FIG. **2**), and whereby both the outermost cover sidewall **1262** of the covering body **12** and the outermost base sidewall **1682** of the base seat **16** are respectively formed with flattened surfaces extended toward the same direction. After the covering body **12**, the sleeving unit **14** and the base seat **16** all are assembled together into the entire stick-shaped paper container **30** shown in FIG. **4A**, the covering body **12** and the base seat **16** both are disposed in a reverse-position manner with relative to each other and are rendered as two truncated-cones stacked from up to down;
- (2) Referring to FIGS. **5A-5C** which depict the third preferred embodiment, the innermost seat sidewall **1684** of the base seat **16** is divided downwardly, from the abutting terminal **164** along the longitudinal centre line **G1**, into an annularly-widened inner diameter section **1686**, an annularly-narrowed inner diameter section **1688**, and an inner circular saddle **1689** formed on a boundary between the annularly-widened inner diameter section **1686** and the annularly-narrowed inner diameter section **1688**. After the abutting terminal **164** of the base seat **16** is disposed with the second end **144** of the sleeving unit **14** (which has a second opening **1442** provided for the content **100** longitudinally passing through), an inner diameter of the receiving opening **1642** of the abutting terminal **164** of the base seat **16** is larger than an outer diameter of the outermost sleeve sidewall **1462** of the sleeving unit **14** such that the second end **144** of the sleeving unit **14** can pass through the receiving opening **1642** of the base seat **16** to enter inside the hollowed base chamber **166** of the base seat **16** until the second end **144** is blocked on the inner circular saddle **1689** from moving to continuously enter inside the annularly-narrowed inner diameter section **1688** since an outer diameter the second end **144** of the sleeving unit **14** is larger than an inner diameter of the annularly-narrowed inner diameter sec-

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tion **1688** but is equal to an inner diameter of the inner circular saddle **1689** of the base seat **16**. Simultaneously, both the annularly-widened inner diameter section **1686** of the base seat **16** and the outermost sleeve sidewall **1462** of the sleeving unit **14** are respectively formed with flattened surfaces longitudinally extended in such a way of constituting a plane-to-plane tight contact in a mutually resting-on manner between the annularly-widened inner diameter section **1686** and the outermost sleeve sidewall **1462** (or adding other permanently retaining structure such as a glue into therebetween). A height of the standing base wall **168** is smaller than a height of the standing sleeve wall **146**, whereby the first end **142** of the sleeving unit **14** can be positioned beyond and outside the abutting terminal **164** of the base seat **16**, in a fitting-on manner between the covering body **12** and the sleeving unit **14**, wherein the first end **142** and an upper portion of the content **100** both are totally sheathed by the hollowed cover chamber **128**; and

- (3) Further referring to FIG. **5B**, a transversally cross-sectional thickness of the standing cover wall **126** of the covering body **12** is substantially equal to an transversally cross-sectional thickness of the annularly-widened inner diameter section **1686** of the standing base wall **168**, but the transversally cross-sectional thickness of the annularly-narrowed inner diameter section **1688** of the standing base wall **168** is larger than the transversally cross-sectional thickness of each of both the standing sleeve wall **146** and the standing cover wall **126**.

Except for the afore-mentioned different structures, the other respective remaining structures of each of the covering body **12**, the sleeving unit **14** and the base seat **16** of the stick-shaped paper container **30** according to the third preferred embodiment shown in FIGS. **5A-5C** are substantially the same as the stick-shaped paper container **20** according to the second preferred embodiment shown in FIGS. **3A-3C** in both preparations and allocations thereof, and therefore all can be referred to the afore-mentioned introductions for the respective components of the stick-shaped paper container **20** according to the second preferred embodiment shown in FIGS. **4A-4C**, but therefore will be omitted in detail later. For example, as shown in FIG. **5B**, after the covering body **12**, the sleeving unit **14** and the base seat **16** all are assembled together into the entire stick-shaped paper container **30**, the stick-shaped paper container **30** has a maximum container height $h2$ parallel to the longitudinal centre line **G1**, a maximum container width $w2$ perpendicular to the longitudinal centre line **G1** and a container height-to-width ratio $r2$ of the maximum container height $h2$ being relative to the maximum container width $w2$; consistently, the container height-to-width ratio $r2$ is between $2^{\sim}5.5$ (namely $h2/w2=r2$, $5.5 \geq r2 \geq 2$). Preferably, while the container height-to-width ratio $r2$ is between $2^{\sim}5.5$, the maximum container width $w2$ is between 12~23 mm.

Nevertheless, it should be noted that in another embodiment based on an improvement to the third preferred embodiment, the stick-shaped paper container **30** could be assembled with usage of only both the covering body **12** and the base seat **16** but without the need of using the sleeving unit **14**, and the covering body **12** and the base seat **16** both are substantially the same in their entire structure and dimension, wherein a lower portion of the content **100** could be permanently retained within the base seat **16** as well as an upper portion of the content **100** is sheathed in an encapsulation manner by the covering body **12**, and thereby consti-

tuting a plane-to-plane loose fit in a manner that the innermost cover sidewall 1282 of the covering body 12 directly props up the upper portion of the content 100.

Furthermore, please refer to the illustrations shown in FIGS. 2 & 6A-6C. FIG. 6A depicts a perspective diagram of a stick-shaped paper container 40 according to a fourth preferred embodiment of the present invention. In the fourth preferred embodiment, the stick-shaped paper container 40 has the assembled components which comprises only both a covering body 12 and a sleeving unit 14. However, the detailed structure and deployed arrangement of each of the covering body 12 and the sleeving unit 14 of the stick-shaped paper container 40 according to the fourth preferred embodiment shown in FIGS. 6A-6C have the following differences, from the covering body 12 and the sleeving unit 14 of the stick-shaped paper container 30 according to the third preferred embodiment shown in FIGS. 5A-5C, that:

- (1) Referring to FIGS. 6A-6C which depict the fourth preferred embodiment, the stick-shaped paper container 40 are assembled into with use of only both the covering body 12 and the sleeving unit 14 but without the need of using the base seat 16 depicted in FIG. 5A. Accordingly, an intermediate portion of the content 100 (such as a recess 109) can be permanently retained in snap-fit by a circumferential wedge formed on the first opening 1422 of the first end 142 of the sleeving unit 14, for restricting the content 100 from movements toward different multi-directions, as well as an upper portion of the content 100 can upwardly pass through the first opening 1422 to be sheathed in an encapsulation manner by the hollowed cover chamber 128 of the covering body 12;
- (2) Further referring to FIGS. 2 & 6A-6C which depict the fourth preferred embodiment, the covering body 12 and the sleeving unit 14 both are respectively fabricated, only by the consistently automatic production machines 70 allocated in the production line used with the improved wet-fiber pulp-molding process (as depicted in FIG. 2) where the first male mold 84 is used to dredge up the wet plant-fibrous pulps 81 and then the first male mold 84 and the first female mold 82 both are mutually matched for compression-molding from the wet plant-fibrous pulps into an integral. This leads the corresponding demold surfaces of the respective entire structure of both the covering body 12 and the sleeving unit 14 to be formed with positive draft angles $\theta 2$ (i.e. smaller than or equal to 2 degrees) relative to the longitudinal centre line G1 (parallel to the longitudinally mold-matching centre line Y1 depicted in FIG. 2). Both the outermost cover sidewall 1262 of the covering body 12 and the outermost sleeve sidewall 1462 of the sleeving unit 14 both are respectively formed with smoothly curved surfaces which are outwardly extended in the same oblique angle, to terminate in both the fitting-on distal end 124 and the second end 144. In the fourth preferred embodiment, after integrally compression-molding used with the wet-fiber pulp-molding process, the outermost cover sidewall 1262 and its corresponding innermost cover sidewall 1282 both are respectively formed with a conical-frustum surface, and the outermost sleeve sidewall 1462 and its corresponding innermost sleeve sidewall 1482 both are formed with a truncated-cone surface, but their shapes are not therefore limited thereto;
- (3) Further referring to FIGS. 6A-6C which depict the fourth preferred embodiment, after the covering body 12 and the sleeving unit 14 both are assembled together,

the covering body 12 and the sleeving unit 14 both can still be respectively rendered in the corresponding positive draft angles $\theta 2$ if the longitudinal centre line G1 is treated as a datum line relative to each position. Accordingly, the sleeving unit 14 can be totally accommodated/nested inside the hollowed cover chamber 128 of the covering body 12; and

- (4) Referring to FIGS. 6A-6C which depict the fourth preferred embodiment, the innermost cover sidewall 1282 is downwardly divided, from the top distal end 122 along the longitudinal centre line G1, into an annularly-narrowed inner diameter section 1284 having a conical surface, an annularly-widened inner diameter section 1286, and a first outer circular saddle 1288 formed between an end of the annularly-widened inner diameter section 1286 and the fitting-on distal end 124. The first outer circular saddle 1288 is used as a transversally-extended stopper. The outermost sleeve sidewall 1462 of the sleeving unit 14 is formed with a smoothly curved surface outwardly extended and has a second outer circular saddle 1488 treated as a transversally-extended stopper.

While the fitting-on distal end 124 of the covering body 12 is disposed with the second end 144 of the sleeving unit 14, the first end 142 of the sleeving unit 12 passes through the fitting-on opening 1242 to enter inside the hollowed cover chamber 128, wherein the smoothly curved surfaces of both the annularly-widened inner diameter section 1286 and the outermost sleeve sidewall 1462 are respectively extended outwardly in the same oblique to terminate in the first and second outer circular saddles 1288, 1488 both which are transversally extended in such a way of constituting a plane-to-plane tight contact (i.e. a loose fit) in a mutually resting-on manner between the annularly-widened inner diameter section 1286 and the outermost sleeve sidewall 1462 (containing the first and second outer circular saddles 1288, 1488), and an outer diameter of the second end 144 is larger than an inner diameter of the fitting-on opening 1242, both the second end 144 and the second outer circular saddle 1488 can be blocked on a bottom side of the first outer circular saddle 1288 from inwardly moving, namely the bottom side of the first outer circular saddle 1288 will longitudinally rest on the second outer circular saddle 1488.

Except for the afore-mentioned different structures, the other respective remaining structures of each of the covering body 12 and the sleeving unit 14 of the stick-shaped paper container 40 according to the fourth preferred embodiment shown in FIGS. 6A-6C are substantially the same as the stick-shaped paper container 30 according to the third preferred embodiment shown in FIGS. 5A-5C, and all can be referred to the afore-mentioned introductions for the respective components of the stick-shaped paper container 30 according to the third preferred embodiment shown in FIGS. 5A-5C, but will be omitted in detail later. For example, as shown in FIG. 6B, when only both the covering body 12 and the sleeving unit 14 are assembled together into the entire stick-shaped paper container 40, the stick-shaped paper container 40 has a maximum container height $h2$ parallel to the longitudinal centre line G1, a maximum container width $w2$ perpendicular to the longitudinal centre line G1, and a container height-to-width ratio $r2$ of the maximum container height $h2$ being relative to the maximum container width $w2$, wherein the container height-to-width ratio $r2$ is between $2^{-3.5}$ (namely $h2/w2=r2$, $3.5 \geq r2 \geq 2$). Preferably, while the container height-to-width ratio $r2$ is between $2^{-3.5}$, the maximum container width $w2$ is between $12 \sim 23$ mm.

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Furthermore, please refer to the illustrations shown in FIGS. 7A-7D, wherein FIG. 7A depicts a perspective diagram of a stick-shaped paper container 50 according to a fifth preferred embodiment of the present invention. The stick-shaped paper container 50 according to the fifth preferred embodiment has the similar assembled components, as afore-mentioned, comprising a covering body 12, a sleeving unit 14 and a base seat 16. Nevertheless, the detailed structure and deployed arrangement of each of the covering body 12, the sleeving unit 14 and the base seat 16 of the stick-shaped paper container 50 according to the fifth preferred embodiment shown in FIGS. 7A-7) has the following differences, from the stick-shaped paper container 10 of the first preferred embodiment shown in FIGS. 3A-3C, that:

(1) Referring to FIGS. 7A-7D which depict the fifth preferred embodiment, the sleeving unit 14 comprises an inner and outer sleeve 14', 14" both of which are nested/stacked together, along the longitudinal centre line G1, to be assembled into the sleeving unit 14. The inner and outer sleeves 14', 14" are respectively formed with two outermost sleeve sidewalls 1462', 1462" both having flattened surfaces longitudinally extended, thereby tightly assembling the inner and outer sleeves 14', 14" together by way of an interlocking retention between a couple of interlocking components 147', 147" respectively formed on the inner and outer sleeves 14', 14". The outermost sleeve sidewall 1462' of the inner sleeve 14' has an end from which an outer circular saddle 1488' is transversally and outwardly extended to terminate in a second end 144'. In the embodiment, the interlocking component 147' of the inner sleeve 14' is a ">"-shaped grooved structure formed along a curve surface of the standing sleeve wall 146'. The interlocking component 147" of the outer sleeve 14" is a hook structure inwardly protruded from the standing sleeve wall 146". While the inner and outer sleeves 14', 14" are nested together in assembly, a first end 142' of the inner sleeve 14' enters inside the hollowed sleeve chamber 148" of the outer sleeve 14" to position the hook structure 147" inside the ">"-shaped grooved structure 147' at a starting end thereof, and next, the outer sleeve 14" can be rotated, in a specific degree relative to the longitudinal centre line G1 (as a center axis) of the inner sleeve 14', to direct the hook structure 147' at rotatably moving from up to down by resting on along an extending passageway of the ">"-shaped grooved structure 147". Finally, the interlocking retention between the coupled interlocking components 147', 147" can be established in such a way of constituting a plane-to-plane tight contact (i.e. a loose fit) in a mutually-and-longitudinally resting-on manner between the outermost sleeve sidewall 1462' of the inner sleeve 14' and the innermost sleeve sidewall 1482' of the outer sleeve 14", whereby the second end 144" of the outer sleeve 14" is blocked and rests on a top side of the outer circular saddle 1488' of the outer sleeve 14'. In another preferred embodiment, the sleeving unit 14 can also be assembled with more than two sleeves. The rust ends 142', 142" of the inner and outer sleeves 14', 14" are respectively formed with two first openings 1422', 1422" such that after the inner and outer sleeves 14', 14" are nested in assembly, the first openings 1422', 1422" provided for the content 100 passing therethrough have respective centre points collinear on the longitudinal centre line G1, whereby the

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respective circumferential wedges of the first openings 1422', 1422" can restrict the content 100 from transversal movement;

(2) Referring to FIGS. 2 & 7A-7D which depict the fifth preferred embodiment, the covering body 12, the inner and outer sleeves 14', 14" (as the sleeving unit 14), and the base seat 16 of the stick-shaped paper container 50 all are integrally molded, respectively, only by way of the preparations of the consistently automatic production machines 70 allocated in the production line used with the improved wet-fiber pulp-molding process (as depicted in FIG. 2) where the first male mold 84 is used to dredge up the wet plant-fibrous pulps 81 and then the first male mold 84 and the first female mold 82 both are mutually matched for compression-molding from the wet plant-fibrous pulps 81 into an integral. After integrally compression-molding, all corresponding demold surfaces of the respective entire structure of the covering body 12, the inner and outer sleeves 14', 14" and the base seat 16 are formed with positive draft angles $\theta 1$ (i.e. smaller than or equal to 1.5 degrees) relative to the longitudinal centre line G1 (parallel to the longitudinally mold-matching centre line Y1 depicted in FIG. 2), and the outermost cover sidewall 1262 of the covering body 12, the two outermost sleeve sidewalls 1462', 1462" of the sleeving unit 14, and the outermost base sidewall 1682 of the base seat 16 all have flattened surfaces longitudinally extended. Preferably, after the integrally compression-molding, the corresponding standing sleeve wall 146', 146" of the inner and outer sleeves 14', 14" are respectively machined by a punching or extrusion process, so as to form the coupled interlocking components 147', 147" thereon, rather than being integrally compression-molded with such coupled interlocking components 147', 147" both rendered in negative draft angles.

(3) Referring to FIGS. 7A-7D which depict the fifth preferred embodiment, the outermost base sidewall 1682 of the base seat 16 is formed with a flattened surface longitudinally extended to reach at an end from which an outer circular saddle 1683 is further transversally and outwardly extended to terminate in a bottom terminal 162. The outer circular saddle 1683 is as a transversally-extended stopper. While the abutting terminal 164 of the base seat 16 is disposed with the second end 144 of the sleeving unit 14', the abutting terminal 164 passes through the second opening 1442 of the sleeving unit 14' to enter inside the hollowed sleeve chamber 148'. The innermost sleeve sidewall 1482' of the inner sleeve 14' and the outermost base sidewall 1682 both are respectively formed with flattened surfaces longitudinally extended, thereby being capable of constituting a plane-to-plane tight contact in a mutually-and-longitudinally resting-on manner between the innermost sleeve sidewall 1482' and the outermost base sidewall 1684 (or adding other permanently retaining structure such as a glue into therebetween). An outer diameter of the abutting terminal 164 is smaller than an inner diameter of each of both the second opening 1442' and the fitting-on opening 1242 such that the abutting terminal 164 can enter inside the hollowed sleeve chamber 148'; however, an outer diameter of the bottom terminal 162 of the base seat 16 is larger than the inner diameter of each of both the second opening 1442' and the fitting-on opening 1242, such that the bottom terminal 162 and the outer circular saddle 1683 both can be blocked on a bottom side of the

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outer circular saddle **1488'** from moving to continuously enter inside the hollowed sleeve chamber **148'**, and the bottom side of the outer circular saddle **1488'** can rest over the outer circular saddle **1683**;

(4) Referring to FIGS. 7A-7) which depicts the fifth preferred embodiment, while the fitting-on distal end **124** of the covering body **12** is disposed over the sleeving unit **14**, the first end **142** of the sleeving unit **14"** passes through the fitting-on opening **1242** of the covering body **12** to enter inside the hollowed cover chamber **128**, wherein the innermost cover sidewall **1282** of the covering body **12** and the outermost sleeve sidewall **1462** of the sleeving unit **14"** both are respectively formed with flatted surfaces longitudinally extended in such a way of constituting a plane-to-plane tight contact (i.e. a loose fit) in a mutually-and-longitudinally resting-on manner between the innermost cover sidewall **1282** and the outermost sleeve sidewall **1462"**. The inner diameter of the fitting-on opening **1242** is larger than an outer diameter of the outermost sleeve sidewall **1462"** such that the entire outermost sleeve sidewall **1462"** of the sleeving unit **14** can enter inside the hollowed cover chamber **128** until the fitting-on distal end **124** of the covering body **12** is blocked on or rest on a top side of the outer circular saddle **1488'**; and

(5) Referring to FIGS. 7A-7D which depict the fifth preferred embodiment, after the covering body **12**, the sleeving unit **14** and the base seat **16** all are assembled together into the entire stick-shaped paper container **50**, the covering body **12**, the sleeving unit **14** and the base seat **16** all still are respectively rendered in the corresponding positive draft angles $\theta 1$ if the longitudinal centre line G1 is treated as a datum line relative to each position; and, the sleeving unit **14** and the base seat **16** both can be accommodated/nested within the hollowed cover chamber **128** of the covering body **12**. Similarly, the stick-shaped paper container **50** has a maximum container height $h2$ parallel to the longitudinal centre line G1, a maximum container width $w2$ perpendicular to the longitudinal centre line (G and a container height-to-width ratio $r2$ of the maximum container height $h2$ being relative to the maximum container width $w2$; and consistently, the container height-to-width ratio $r2$ is between $2 \sim 3.5$ (namely $h2/w2=r2$, $5.5 \geq r2 \geq 2$). Preferably, while the container height-to-width ratio $r2$ is between $2 \sim 3.5$, the maximum container width $w2$ is between $12 \sim 23$ mm.

Except for the afore-mentioned different structures, the other respective remaining structures of the covering body **12**, the sleeving unit **14** and the base seat **16** of the stick-shaped paper container **50** of the fifth preferred embodiment shown in FIGS. 7A-7C all are substantially the same as the stick-shaped paper container **10** of the first preferred embodiment shown in FIGS. 3A-3C in their preparation and allocations, and all can be referred to the afore-mentioned introductions for the respective components of the stick-shaped paper container **10** of the first preferred embodiment shown in FIGS. 3A-3C, but therefore will be omitted in detail later.

Nevertheless, in another preferred embodiment, the respective transversally cross-sectional thicknesses of all the covering body **12**, the two inner and outer sleeves **14'**, **14"** and the base seat **16** are different from each other. It should be noted that, for another embodiment on a basis of an improvement to the fifth preferred embodiment, the stick-shaped paper container **50** can be assembled with the use of

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only both the covering body **12** and the sleeving unit **14**, but without the need of using the base seat **16**, such that an intermediate portion of or a lower portion of the content **100** can be permanently retained within the first openings **1422'**, **1422"** of the sleeving unit **14**, and an upper portion of the content **100** can be sheathed, in an encapsulation manner, within the hollowed cover chamber **128** of the covering body **12**.

Compared with the conventional arts, each of all the components of the stick-shaped paper container **10**, **20**, **30**, **40**, **50**, according to the present invention, is capable of being integrally molded by way of the preparations of the consistently automatic production machines **70** allocated in the production line used with the improved wet-fiber pulp-molding process (as depicted in FIG. 2). This does not only resolve said technical drawback of the conventional pulp-molding process which is incapable of producing such a paper article having a height-to-width ratio, greater than one, of its maximum longitudinal height being relative to its maximum transversal width, but also reduces its working cycle time, benefits its mass production, and assures its higher product yield and quality. Furthermore, during a compression-molding used with the wet-fiber pulp-molding process, the respective demold surfaces of all the components of the stick-shaped paper containers **10**, **20**, **30**, **40**, **50** according to the present invention all are rendered in the corresponding positive draft angles relative to the longitudinal demolding direction Y1 (See FIG. 2) in the series of mold assemblies **82**, **84**, **92**, **94**, rather than the negative draft angles as rendered in the respective demold surfaces of the laterally latch structure and/or laterally recessed structure of the conventional pulp-molded article. Consequently, it is unnecessary to additionally dispose any other transversal mold implemented for horizontally demolding, and/or a slidable block device implemented for horizontal movement, along with collocating the mold assemblies for vertically demolding, and thereby being capable of simplifying the mold assemblies, and lowering its mold cost.

As described above, although the present invention has been described with the preferred embodiments thereof, those skilled in the art will appreciate that various modifications, additions, and substitutions are possible without departing from the scope and the spirit of the invention. Accordingly, the scope of the present invention is intended to be defined only by reference to the claims.

What is claimed is:

1. A container, comprising:

a sleeving unit, including:

a first opening;

a first section having a first diameter; and

a second opening having a second diameter, wherein the second diameter is larger than the first diameter;

a covering body configured to removably retain the first section of the sleeving unit with friction; and

a base seat removably coupled to the sleeving unit, wherein an outermost base sidewall of the base seat is retained in the second opening of the sleeving unit with friction wherein

the sleeving unit, the covering body, and the base seat comprise plant fiber.

2. The container of claim 1, wherein the covering body is conical.

3. The container of claim 1, wherein the sleeving unit is tapered from the second opening to the first opening.

4. The container of claim 1, wherein the sleeving unit further comprises a saddle below the first section.

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5. The container of claim 4, wherein the covering body contacts the saddle when the first section of the sleeving unit is retained in the covering body.

6. The container of claim 5, wherein the third diameter is larger than the first diameter. 5

7. The container of claim 5, wherein the third diameter is substantially equal to the second diameter.

8. The container of claim 4, wherein the saddle defines a third diameter.

9. The container of claim 1, wherein the base seat further 10
comprises a bottom terminal.

10. The container of claim 9, wherein the bottom terminal is flush with a standing sleeve wall of the sleeving unit.

11. The container of claim 1, wherein the base seat further 15
comprises an outer circular saddle.

12. The container of claim 1, wherein the container further comprises an adhesive located between the base seat and the sleeving unit.

13. The container of claim 1, wherein the container is 20
configured to retain a cosmetic.

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